

Safety Data Sheet

SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**1.1 Product Identifier**

Material Name : Shell Ondina Oil 934
Product Code : 901L2685
CAS No. : 8042-47-5
REACH Registration No. : 01-2119487078-27-0005

1.2 Relevant identified uses of the substance or mixture and uses advised against

Product Use : Process oil. Please refer to Ch16 for the registered uses under REACH.

Uses Advised Against : This product must not be used in applications other than those recommended in Section 1, without first seeking the advice of the supplier.

1.3 Details of the Supplier of the safety data sheet

Manufacturer/Supplier : Shell UK Oil Products Limited
Shell Centre
London
SE1 7NA
United Kingdom

Telephone : (+44) 08007318888

Email Contact for Safety Data Sheet : If you have any enquiries about the content of this SDS please email lubricantSDS@shell.com

1.4 Emergency Telephone Number

: +44-(0) 151-350-4595

SECTION 2. HAZARDS IDENTIFICATION**2.1 Classification of the substance or mixture**

Regulation (EC) No 1272/2008 (CLP)	
Hazard classes / Hazard categories	Hazard Statement
Not classified	

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67/548/EEC or 1999/45/EC	
Hazard Characteristics	R-phrases(s)
Not classified as dangerous under EC criteria.;	

2.2 Label Elements**Labeling according to Regulation (EC) No 1272/2008****Hazard pictograms**

: No symbol

Signal Words

: No signal word

CLP Hazard Statements

: PHYSICAL HAZARDS:
Not classified as a physical hazard under GHS criteria.

HEALTH HAZARDS:
Not classified as a health hazard under GHS criteria.

ENVIRONMENTAL HAZARDS:
Not classified as an environmental hazard under GHS criteria.

CLP Precautionary statements**Prevention**

: No precautionary phrases.

Response

: No precautionary phrases.

Storage

: No precautionary phrases.

Disposal:

: No precautionary phrases.

2.3 Other Hazards

: Not classified as flammable but will burn.

Prolonged or repeated skin contact without proper cleaning can clog the pores of the skin resulting in disorders such as oil acne/folliculitis. Used oil may contain harmful impurities.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**3.1 Substance**

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Material Name : Highly refined mineral oil.

CAS No. : 8042-47-5

3.2 Mixtures

Mixture Description : Product is not a mixture according to regulation 1907/2006/EC.

Additional Information : The highly refined mineral oil contains <3% (w/w) DMSO-extract, according to IP346.

SECTION 4. FIRST-AID MEASURES

4.1 Description of First Aid Measures

General Information : Not expected to be a health hazard when used under normal conditions.

Inhalation : No treatment necessary under normal conditions of use. If symptoms persist, obtain medical advice.

Skin Contact : Remove contaminated clothing. Flush exposed area with water and follow by washing with soap if available. If persistent irritation occurs, obtain medical attention.

Eye Contact : Flush eye with copious quantities of water. If persistent irritation occurs, obtain medical attention.

Ingestion : If vomiting occurs spontaneously, keep head below hips to prevent aspiration. Give nothing by mouth.

Self-protection of the first aider : When administering first aid, ensure that you are wearing the appropriate personal protective equipment according to the incident, injury and surroundings.

4.2 Most important symptoms and effects, both acute and delayed : Oil acne/folliculitis signs and symptoms may include formation of black pustules and spots on the skin of exposed areas. Ingestion may result in nausea, vomiting and/or diarrhoea.

4.3 Indication of any immediate medical attention and special treatment needed : Notes to doctor/physician:
Treat symptomatically.

SECTION 5. FIRE-FIGHTING MEASURES

Clear fire area of all non-emergency personnel.

5.1 Extinguishing Media : Foam, water spray or fog. Dry chemical powder, carbon

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- Unsuitable Extinguishing Media** : dioxide, sand or earth may be used for small fires only.
Do not use water in a jet.
- 5.2 Special hazards arising from the substance or mixture** : Hazardous combustion products may include: A complex mixture of airborne solid and liquid particulates and gases (smoke). Carbon monoxide may be evolved if incomplete combustion occurs. Unidentified organic and inorganic compounds.
- 5.3 Advice for firefighters** : Proper protective equipment including chemical resistant gloves are to be worn; chemical resistant suit is indicated if large contact with spilled product is expected. Self-Contained Breathing Apparatus must be worn when approaching a fire in a confined space. Select fire fighter's clothing approved to relevant Standards (e.g. Europe: EN469).

SECTION 6. ACCIDENTAL RELEASE MEASURES

Avoid contact with spilled or released material. For guidance on selection of personal protective equipment see Chapter 8 of this Material Safety Data Sheet. Observe the relevant local and international regulations.

- 6.1 Personal Precautions, Protective Equipment and Emergency Procedures** : 6.1.1 For non emergency personnel: Avoid contact with skin and eyes.
6.1.2 For emergency responders: Avoid contact with skin and eyes.
- 6.2 Environmental Precautions** : Use appropriate containment to avoid environmental contamination. Prevent from spreading or entering drains, ditches or rivers by using sand, earth, or other appropriate barriers.
- 6.3 Methods and Material for Containment and Cleaning Up** : Slippery when spilt. Avoid accidents, clean up immediately. Prevent from spreading by making a barrier with sand, earth or other containment material. Reclaim liquid directly or in an absorbent. Soak up residue with an absorbent such as clay, sand or other suitable material and dispose of properly.
- Additional Advice** : Local authorities should be advised if significant spillages cannot be contained.
- 6.4 Reference to other sections** : For guidance on selection of personal protective equipment see Chapter 8 of this Material Safety Data Sheet. For guidance on disposal of spilled material see Chapter 13 of this Material Safety Data Sheet.

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SECTION 7. HANDLING AND STORAGE

- General Precautions** : Use local exhaust ventilation if there is risk of inhalation of vapours, mists or aerosols. Use the information in this data sheet as input to a risk assessment of local circumstances to help determine appropriate controls for safe handling, storage and disposal of this material.
- 7.1 Precautions for Safe Handling** : Avoid prolonged or repeated contact with skin. Avoid inhaling vapour and/or mists. When handling product in drums, safety footwear should be worn and proper handling equipment should be used. Properly dispose of any contaminated rags or cleaning materials in order to prevent fires. Keep container tightly closed and in a cool, well-ventilated place. Use properly labelled and closeable containers.
- Product Transfer** : This material has the potential to be a static accumulator. Proper grounding and bonding procedures should be used during all bulk transfer operations.
- 7.2 Conditions for safe storage, including any incompatibilities** : Store at ambient temperature.
- Store separately from oxidising agents.
Refer to section 15 for any additional specific legislation covering the packaging and storage of this product.
The storage of this product may be subject to the Control of Pollution (Oil Storage) (England) Regulations. Further guidance may be obtained from the local environmental agency office.
- Recommended Materials** : For containers or container linings, use mild steel or high density polyethylene.
- Unsuitable Materials** : PVC.
- 7.3 Specific end use(s)** : Not applicable
- Additional Information** : Polyethylene containers should not be exposed to high temperatures because of possible risk of distortion. Exposure to this product should be reduced as low as reasonably practicable. Reference should be made to the Health and Safety Executive's publication "COSHH Essentials".

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

If the American Conference of Governmental Industrial Hygienists (ACGIH) value is provided on this document, it is provided for information only.

8.1 Control Parameters

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Occupational Exposure Limits

Material	Source	Type	ppm	mg/m3	Notation
Oil mist, mineral	ACGIH	TWA(Inhalable fraction.)		5 mg/m3	

Biological Exposure Index (BEI)

No biological limit allocated.

PNEC related information : Substance is a hydrocarbon with a complex, unknown or variable composition. Conventional methods of deriving PNECs are not appropriate and it is not possible to identify a single representative PNEC for such substances.

Monitoring Methods : Monitoring of the concentration of substances in the breathing zone of workers or in the general workplace may be required to confirm compliance with an OEL and adequacy of exposure controls. For some substances biological monitoring may also be appropriate. Validated exposure measurement methods should be applied by a competent person and samples analysed by an accredited laboratory. Examples of sources of recommended exposure measurement methods are given below or contact the supplier. Further national methods may be available.

National Institute of Occupational Safety and Health (NIOSH), USA: Manual of Analytical Methods <http://www.cdc.gov/niosh/>

Occupational Safety and Health Administration (OSHA), USA: Sampling and Analytical Methods <http://www.osha.gov/>

Health and Safety Executive (HSE), UK: Methods for the Determination of Hazardous Substances
<http://www.hse.gov.uk/>

Institut für Arbeitsschutz Deutschen Gesetzlichen Unfallversicherung (IFA), Germany.
<http://www.dguv.de/inhalt/index.jsp>

L'Institut National de Recherche et de Sécurité, (INRS), France

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<http://www.inrs.fr/accueil>

**8.2 Exposure Controls
General Information**

: The level of protection and types of controls necessary will vary depending upon potential exposure conditions. Select controls based on a risk assessment of local circumstances. Appropriate measures include: Adequate ventilation to control airborne concentrations. Where material is heated, sprayed or mist formed, there is greater potential for airborne concentrations to be generated.

Define procedures for safe handling and maintenance of controls. Educate and train workers in the hazards and control measures relevant to normal activities associated with this product. Ensure appropriate selection, testing and maintenance of equipment used to control exposure, e.g. personal protective equipment, local exhaust ventilation. Drain down system prior to equipment break-in or maintenance. Retain drain downs in sealed storage pending disposal or for subsequent recycle. Always observe good personal hygiene measures, such as washing hands after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants. Discard contaminated clothing and footwear that cannot be cleaned. Practice good housekeeping.

Occupational Exposure Controls**Personal Protective
Equipment**

: The provided information is made in consideration of the PPE directive (Council Directive 89/686/EEC) and the CEN European Committee for Standardisation (CEN) standards. Personal protective equipment (PPE) should meet recommended national standards. Check with PPE suppliers.

Eye Protection

: Wear safety glasses or full face shield if splashes are likely to occur. Approved to EU Standard EN166.

Hand Protection

: Where hand contact with the product may occur the use of gloves approved to relevant standards (e.g. Europe: EN374, US: F739) made from the following materials may provide suitable chemical protection: PVC, neoprene or nitrile rubber gloves. Suitability and durability of a glove is dependent on usage, e.g. frequency and duration of contact, chemical resistance of glove material, dexterity. Always seek advice from glove suppliers. Contaminated gloves should be replaced.

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Personal hygiene is a key element of effective hand care. Gloves must only be worn on clean hands. After using gloves, hands should be washed and dried thoroughly. Application of a non-perfumed moisturizer is recommended.

For continuous contact we recommend gloves with breakthrough time of more than 240 minutes with preference for > 480 minutes where suitable gloves can be identified. For short-term/splash protection we recommend the same, but recognise that suitable gloves offering this level of protection may not be available and in this case a lower breakthrough time may be acceptable so long as appropriate maintenance and replacement regimes are followed. Glove thickness is not a good predictor of glove resistance to a chemical as it is dependent on the exact composition of the glove material. Glove thickness should be typically greater than 0.35 mm depending on the glove make and model.

Body protection : Skin protection not ordinarily required beyond standard issue work clothes.

Respiratory Protection : No respiratory protection is ordinarily required under normal conditions of use. In accordance with good industrial hygiene practices, precautions should be taken to avoid breathing of material. If engineering controls do not maintain airborne concentrations to a level which is adequate to protect worker health, select respiratory protection equipment suitable for the specific conditions of use and meeting relevant legislation. Check with respiratory protective equipment suppliers. Where air-filtering respirators are suitable, select an appropriate combination of mask and filter. Select a filter suitable for combined particulate/organic gases and vapors [Type A/Type P boiling point > 65°C (149°F)] meeting EN14387 and EN143.

Thermal Hazards : Not applicable.

Environmental Exposure Controls

Environmental exposure control measures : Local guidelines on emission limits for volatile substances must be observed for the discharge of exhaust air containing vapour. Take appropriate measures to fulfil the requirements of relevant environmental protection legislation. Avoid contamination of the environment by following advice given in Chapter 6. If necessary, prevent undissolved material from being discharged to waste water. Waste water should be treated in a municipal or industrial waste water treatment plant before discharge to surface water.

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SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**9.1 Information on basic physical and chemical properties**

Appearance	: Clear colourless. Liquid at room temperature.
Odour	: Odourless.
Odour threshold	: Data not available
pH	: Not applicable.
Initial Boiling Point and Boiling Range	: > 280 °C / 536 °F estimated value(s)
Pour point	: Typical -12 °C / 10 °F
Flash point	: Typical 240 °C / 464 °F (COC)
Upper / lower Flammability or Explosion limits	: Typical 1 - 10 %(V) (based on mineral oil)
Auto-ignition temperature	: > 320 °C / 608 °F
Vapour pressure	: < 0.5 Pa at 20 °C / 68 °F (estimated value(s))
Relative Density	: Typical 0.865 at 15 °C / 59 °F
Density	: Typical 865 kg/m ³ at 15 °C / 59 °F
Water solubility	: Negligible.
Solubility in other solvents	: Data not available
n-octanol/water partition coefficient (log Pow)	: > 6 (based on information on similar products)
Dynamic viscosity	: Data not available
Kinematic viscosity	: 65 - 70 mm ² /s at 40 °C / 104 °F
Vapour density (air=1)	: > 1 (estimated value(s))
Evaporation rate (nBuAc=1)	: Data not available
Decomposition	: Data not available
Temperature	
Flammability	: Data not available
Oxidizing Properties	: Data not available
Explosive Properties	: Not classified

9.2 Other Information

Other Information	: not a VOC
Volatile organic compound	: 0 %

SECTION 10. STABILITY AND REACTIVITY

10.1 Reactivity	: The product does not pose any further reactivity hazards in
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- addition to those listed in the following sub-paragraph.
- 10.2 Chemical stability** : No hazardous reaction is expected when handled and stored according to provisions.
- 10.3 Possibility of Hazardous Reactions** : Reacts with strong oxidising agents.
- 10.4 Conditions to Avoid** : Extremes of temperature and direct sunlight.
- 10.5 Incompatible Materials** : Strong oxidising agents.
- 10.6 Hazardous Decomposition Products** : Hazardous decomposition products are not expected to form during normal storage.

SECTION 11. TOXICOLOGICAL INFORMATION

11.1 Information on Toxicological effects

- Basis for Assessment** : Information given is based on data on the components and the toxicology of similar products.
- Likely Routes of Exposure** : Skin and eye contact are the primary routes of exposure although exposure may occur following accidental ingestion.
- Acute Oral Toxicity** : Low toxicity: LD50 > 5000 mg/kg , Rat
- Acute Dermal Toxicity** : Low toxicity: LD50 > 5000 mg/kg , Rabbit
- Acute Inhalation Toxicity** : Low toxicity: LC50 >5 mg/l / 4 h, Rat
- Skin corrosion/irritation** : Not irritating to skin. Prolonged or repeated skin contact without proper cleaning can clog the pores of the skin resulting in disorders such as oil acne/folliculitis.
- Serious eye damage/irritation** : Expected to be non-irritating to eyes.
- Respiratory Irritation** : Inhalation of vapours or mists may cause irritation to the respiratory system.
- Respiratory or skin sensitisation** : For respiratory and skin sensitisation: Not expected to be a sensitiser.
- Aspiration Hazard** : Not considered an aspiration hazard.
- Germ cell mutagenicity** : Not expected to be mutagenic.
- Carcinogenicity** : Not expected to be carcinogenic. Product contains mineral oils of types shown to be non-carcinogenic in animal skin-painting studies. Highly refined mineral oils are not classified as carcinogenic by the International Agency for Research on Cancer (IARC).

Material		Carcinogenicity Classification
Highly refined mineral oil (IP346 <3%)	:	ACGIH Group A4: Not classifiable as a human carcinogen.

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Highly refined mineral oil (IP346 <3%)	:	IARC 3: Not classifiable as to carcinogenicity to humans.
Highly refined mineral oil (IP346 <3%)	:	GHS / CLP: No carcinogenicity classification

Summary on evaluation of the CMR properties

Carcinogenicity	:	This product does not meet the criteria for classification in categories 1A/1B.,
Mutagenicity	:	This product does not meet the criteria for classification in categories 1A/1B.
Reproductive Toxicity (fertility)	:	This product does not meet the criteria for classification in categories 1A/1B.
Specific target organ toxicity - single exposure	:	Not classified.
Specific target organ toxicity - repeated exposure	:	Not expected to be a hazard.
Additional Information	:	Used oils may contain harmful impurities that have accumulated during use. The concentration of such impurities will depend on use and they may present risks to health and the environment on disposal. ALL used oil should be handled with caution and skin contact avoided as far as possible. Classifications by other authorities under varying regulatory frameworks may exist.

SECTION 12. ECOLOGICAL INFORMATION

Basis for Assessment	:	Ecotoxicological data have not been determined specifically for this product. Information given is based on a knowledge of the components and the ecotoxicology of similar products.
12.1 Toxicity		
Acute Toxicity	:	LL/EL50 expressed as the nominal amount of product required to prepare aqueous test extract. Expected to be not toxic at limit of water solubility.
Fish	:	Expected to be practically non toxic: LL/EL/IL50 > 100 mg/l
Aquatic crustacea	:	Expected to be practically non toxic: LL/EL/IL50 > 100 mg/l
Algae/aquatic plants	:	Expected to be practically non toxic: LL/EL/IL50 > 100 mg/l
Microorganisms	:	Expected to be practically non toxic: LL/EL/IL50 > 100 mg/l
Chronic Toxicity		
Fish	:	NOEC/NOEL expected to be > 10 - <= 100 mg/l
Aquatic crustacea	:	NOEC/NOEL expected to be > 10 - <= 100 mg/l

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- 12.2 Persistence and degradability** : Expected to be inherently biodegradable.
- 12.3 Bioaccumulative Potential** : Has the potential to bioaccumulate.
- 12.4 Mobility in Soil** : Liquid under most environmental conditions. If it enters soil, it will adsorb to soil particles and will not be mobile. Floats on water.
- 12.5 Result of PBT and vPvB assesment** : The substance does not fulfill all screening criteria for persistence, bioaccumulation and toxicity and hence is not considered to be PBT or vPvB.
- 12.6 Other Adverse Effects** : Films formed on water may affect oxygen transfer and damage organisms. May cause physical fouling of aquatic organisms.

SECTION 13. DISPOSAL CONSIDERATIONS

13.1 Waste Treatment Methods

- Material Disposal** : Recover or recycle if possible. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste classification and disposal methods in compliance with applicable regulations. Do not dispose into the environment, in drains or in water courses.
- Container Disposal** : Dispose in accordance with prevailing regulations, preferably to a recognised collector or contractor. The competence of the collector or contractor should be established beforehand.
- Local Legislation** : Disposal should be in accordance with applicable regional, national, and local laws and regulations.
EU Waste Disposal Code (EWC): 13 08 99 oil waste not otherwise specified. Classification of waste is always the responsibility of the end user.

SECTION 14. TRANSPORT INFORMATION

Land transport (ADR/RID): ADR

This product is not classified as dangerous for this mode of transport. Therefore 14.1 UN

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Number, 14.2 UN Proper Shipping name, 14.3 Transport hazard class(es), 14.4 Packing group, 14.5 Environmental hazards, 14.6 Special precautions for user do not apply.

RID

This product is not classified as dangerous for this mode of transport. Therefore 14.1 UN Number, 14.2 UN Proper Shipping name, 14.3 Transport hazard class(es), 14.4 Packing group, 14.5 Environmental hazards, 14.6 Special precautions for user do not apply.

Inland waterways transport (ADN):

This product is not classified as dangerous for this mode of transport. Therefore 14.1 UN Number, 14.2 UN Proper Shipping name, 14.3 Transport hazard class(es), 14.4 Packing group, 14.5 Environmental hazards, 14.6 Special precautions for user do not apply.

Sea transport (IMDG Code):

This product is not classified as dangerous for this mode of transport. Therefore 14.1 UN Number, 14.2 UN Proper Shipping name, 14.3 Transport hazard class(es), 14.4 Packing group, 14.5 Environmental hazards, 14.6 Special precautions for user do not apply.

Air transport (IATA):

This product is not classified as dangerous for this mode of transport. Therefore 14.1 UN Number, 14.2 UN Proper Shipping name, 14.3 Transport hazard class(es), 14.4 Packing group, 14.5 Environmental hazards, 14.6 Special precautions for user do not apply.

14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Pollution Category	: Not applicable.
Ship Type	: Not applicable.
Product Name	: Not applicable.
Special Precaution	: Not applicable.

Additional Information : MARPOL Annex 1 rules apply for bulk shipments by sea.

SECTION 15. REGULATORY INFORMATION

The regulatory information is not intended to be comprehensive. Other regulations may apply to this material.

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Other regulatory Information

Authorisations and/or restrictions on use : Product is not subject to Authorisation under REACH.

Recommended Restrictions on Use : This product must not be used in applications other than those recommended in Section 1, without first seeking the advice of

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(Advice Against) the supplier.

Chemical Inventory Status

EINECS : All components
listed or polymer
exempt.

TSCA : All components
listed.

Other Information : Environmental Protection Act 1990 (as amended).
Health and Safety at Work etc. Act 1974.
Consumers Protection Act 1987.
Pollution Prevention and Control Act 1999.
Environment Act 1995.
Factories Act 1961.
The Carriage of Dangerous Goods and Use of Transportable
Pressure Equipment (Amendment) Regulations 2011.
Chemicals (Hazard Information and Packaging for Supply)
Regulations 2009.
Control of Substances Hazardous to Health Regulations 2002
(as amended).
Merchant Shipping (Dangerous Goods and Marine Pollutants)
Regulations 1997.
Reporting of Injuries, Diseases and Dangerous Occurrences
Regulations 1995 (as amended).
Personal Protective Equipment Regulations 2002.
Personal Protective Equipment at Work Regulations 1992.
Hazardous Waste (England and Wales) Regulations 2005(as
amended).
Control of Major Accident Hazards Regulations 1999 (as
amended).
Renewable Transport Fuel Obligations Order 2007 (as
amended).
Energy Act 2011.
Environmental Permitting (England and Wales) Regulations
2010 (as amended).
Waste (England and Wales) Regulations 2011 (as amended).
Planning (Hazardous Substances) Act 1990 and associated
regulations.
The Environmental Protection (Controls on Ozone-Depleting
Substances) Regulations 2011.

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15.2 Chemical Safety Assessment : A Chemical Safety Assessment was performed for this substance.

SECTION 16. OTHER INFORMATION**Identified Uses according to the Use Descriptor System****Uses - Worker**

Title : - Industrial
Manufacture of substance
Distribution of substance
Use as an intermediate
Formulation & (re)packing of substances and mixtures
Uses in Coatings
Use in Cleaning Agents
Lubricants
Metal working fluids / rolling oils
Use as binders and release agents
Functional Fluids
Use in laboratories
Rubber production and processing
Water treatment chemicals
Polymer processing

Uses - Worker

Title : - Professional
Uses in Coatings
Use in Cleaning Agents
Lubricants
Metal working fluids / rolling oils
Use as binders and release agents
Use in Agrochemicals uses
Functional Fluids
Use in laboratories
Water treatment chemicals
Explosives manufacture & use

Uses - Consumer

Title : - Consumer
Uses in Coatings
Use in Cleaning Agents
Lubricants
Use in Agrochemicals uses

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Use as a fuel
Other Consumer Uses

Additional Information : For a list of REACH registered uses, please refer to:
http://www.shell.com/reach_uses
This product is not classified for human health or environmental hazards. An exposure scenario is not required.

Other Information

Abbreviations and Acronyms : The standard abbreviations and acronyms used in this document can be looked up in reference literature (e.g. scientific dictionaries) and/or websites.

ACGIH = American Conference of Governmental Industrial Hygienists
ADR = European Agreement concerning the International Carriage of Dangerous Goods by Road
AICS = Australian Inventory of Chemical Substances
ASTM = American Society for Testing and Materials
BEL = Biological exposure limits
BTEX = Benzene, Toluene, Ethylbenzene, Xylenes
CAS = Chemical Abstracts Service
CEFIC = European Chemical Industry Council
CLP = Classification Packaging and Labelling
COC = Cleveland Open-Cup
DIN = Deutsches Institut für Normung
DMEL = Derived Minimal Effect Level
DNEL = Derived No Effect Level
DSL = Canada Domestic Substance List
EC = European Commission
EC50 = Effective Concentration fifty
ECETOC = European Center on Ecotoxicology and Toxicology Of Chemicals
ECHA = European Chemicals Agency
EINECS = The European Inventory of Existing Commercial Chemical Substances
EL50 = Effective Loading fifty
ENCS = Japanese Existing and New Chemical Substances Inventory
EWC = European Waste Code
GHS = Globally Harmonised System of Classification and Labelling of Chemicals

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IARC = International Agency for Research on Cancer
 IATA = International Air Transport Association
 IC50 = Inhibitory Concentration fifty
 IL50 = Inhibitory Level fifty
 IMDG = International Maritime Dangerous Goods
 INV = Chinese Chemicals Inventory
 IP346 = Institute of Petroleum test method N° 346 for the determination of polycyclic aromatics DMSO-extractables
 KECI = Korea Existing Chemicals Inventory
 LC50 = Lethal Concentration fifty
 LD50 = Lethal Dose fifty per cent.
 LL/EL/IL = Lethal Loading/Effective Loading/Inhibitory loading
 LL50 = Lethal Loading fifty
 MARPOL = International Convention for the Prevention of Pollution From Ships
 NOEC/NOEL = No Observed Effect Concentration / No Observed Effect Level
 OE_HP V = Occupational Exposure - High Production Volume
 PBT = Persistent, Bioaccumulative and Toxic
 PICCS = Philippine Inventory of Chemicals and Chemical Substances
 PNEC = Predicted No Effect Concentration
 REACH = Registration Evaluation And Authorisation Of Chemicals
 RID = Regulations Relating to International Carriage of Dangerous Goods by Rail
 SKIN_DES = Skin Designation
 STEL = Short term exposure limit
 TRA = Targeted Risk Assessment
 TSCA = US Toxic Substances Control Act
 TWA = Time-Weighted Average
 vPvB = very Persistent and very Bioaccumulative

SDS Distribution : The information in this document should be made available to all who may handle the product.
SDS Version Number : 2.1
SDS Effective Date : 11.09.2013
SDS Revisions : A vertical bar (|) in the left margin indicates an amendment from the previous version.
SDS Regulation : Regulation 1907/2006/EC as amended by Regulation (EU) 453/2010
Disclaimer : This information is based on our current knowledge and is

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intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.